

HTML and CSS — Mark schemes

PAPERS A TO D • 30 MARKS EACH • TUTOR COPY

Marks in square brackets show how the total is split: one tagged line per mark. Code is marked on what it does, not on how it is laid out. Always accept: single or double quote marks round an attribute value; capitals in tag and property names; any indentation, or the whole thing on one line; attributes in any order; `<br>` or `<br/>`, and `<img ... />` as well as `<img ...>`. Never accept: a missing quote mark (withhold the mark for that attribute), a closing tag on `img` or `br`, colour or centre in CSS, or a hex code without the #. A missing semicolon after the last declaration is legal CSS and is not penalised; a missing semicolon between two declarations is. Mark the tags and properties, not the spelling of the words between them. Every code listing printed in these papers was run through a parser when the papers were built, so each fault listed here is a real fault and the only one. The notes under each answer name the mistake that question was built to catch.

Paper A — Medium

30 MARKS • 35 MINUTES

- 1** (a) `<a href="about.html">` (the whole opening tag, attribute included) [1] [4]
 (b) **href** [1]
 (c) **about.html**, with or without the quote marks [1]
 (d) **element** [1]
Tag, attribute, element: three words, and the exam uses all three. The usual slip is (d): answering "tag" for the whole piece. A tag is one half; an element is opening tag, content and closing tag together. For (a), "a" or <a> alone loses the mark, because the attribute is part of the opening tag.
- 2** Line 3: `<head>` [1] [4]
 Line 4: `<title>` [1]
 Line 5: `</head>` [1]
 Line 6: `<body>` [1]
Line 5 is the one that catches people: the missing tag is a closing tag, and without the slash it is a different tag. On line 5, "head" or <head> scores nothing. On lines 3, 4 and 6 accept the bare word (head, title, body) if it is clear which tag is meant. Capital letters are fine.
- 3** Model answer: [3]
`<h1>Meera's Bakery</h1>`
`<h2>Opening hours</h2>`
`<p>Open 7 am to 9 pm every day.</p>`
`<h1>Meera's Bakery</h1>` [1]
`<h2>Opening hours</h2>` (h2: a section sitting directly under the h1) [1]
`<p>Open 7 am to 9 pm every day.</p>` [1]
The mistake this was built to catch is choosing a heading by its size. An h3 or h4 for "Opening hours" loses the second mark, however sensible it looks on screen: it skips the outline. Mark the tags, not the spelling of the words between them. A tag opened and never closed loses that line's mark. One line or three, any indentation, capitals: all fine.

- 4 Model answer: [3]
- ```

 Boil water
 Add tea
 Pour

```
- ol** opening and closing tags (not ul) [1]  
 three **li** elements, each opened and closed, holding Boil water, Add tea and Pour in that order [1]  
 only li directly inside the list, and no typed numbers [1]
- Typed numbers ("1. Boil water") lose the third mark: the browser counts for you, which is the whole point of an ordered list. A student who uses ul has built a bulleted list; take the first mark only and give the other two if the li structure is right. Items in the wrong order lose the second mark.*
- 5 Model answer: [3]
- ```

```
- an **img** tag with **no closing tag** (is also accepted) [1]
 src="kulfi.jpg" (src, not href) [1]
 an **alt** that describes the picture: pink kulfi, a vendor, a steel cart [1]
- Three separate traps. or ... loses the first mark. href for src loses the second. For the third, "image", "photo", "kulfi" and "kulfi.jpg" all score nothing: they say nothing the visitor could not guess. Missing alt is the same. Attribute order does not matter; single or double quotes are both fine; no quotes at all loses the mark for that attribute.*
- 6 The browser shows: Doors open at 9. / Match starts at 10. / 1. Toss / 2. Batting / 3. Tea [3]
- the paragraph is on **two lines**, broken after "9." by the br [1]
 the list is **numbered** 1, 2, 3 (not bullets) [1]
 the items read Toss, Batting, Tea in order, with no tag visible anywhere on the screen [1]
- Every wrong answer here comes from reading the code as typed instead of as drawn:
 printed on the screen, bullets for an ol, or the whole paragraph run onto one line. A student who ignores the br still gets the other two marks. Accept written lines or a drawn picture.*
- 7 A: **border** [1] [4]
 B: **content** (accept "text") [1]
 C: **margin** [1]
 D: **padding** [1]
- The arrows are not in nesting order on purpose, so the answers cannot be guessed from the order they are printed in. The common slip is margin and padding swapped (C and D): withhold both and keep A and B. The reminder box gives the order of the layers, so none of this should be a guess.*

- 8** Model answer: [3]
- ```
h2 {
 color: darkblue;
 background-color: yellow;
 text-align: center;
}
```
- selector **h2**, curly braces, and each declaration written as property, colon, value, semicolon [1]  
**color** for the text and **background-color** for the yellow (any dark blue: darkblue, navy, #00008B, rgb(0, 0, 139); plain blue is also accepted) [1]  
 text-align: center; [1]
- British spelling is the target. colour and centre each lose their mark: CSS ignores both without a word of complaint, which is why the mistake survives. Swapping the two colours (color for the yellow) loses the second mark. A missing semicolon after the last declaration is legal CSS and is not penalised; a missing semicolon between two declarations loses the first mark. One line or several, any order, capitals in property names: all fine.*
- 9** (a)  $100 + 2 \times 10 + 2 \times 2 + 2 \times 8$  [1] [3]  
 $= 100 + 20 + 4 + 16 = 140 \text{ px}$  [1]  
 (b) the **border** [1] ( $100 + 20 + 16 = 136$ )
- Forgetting the border is the commonest slip in this topic, which is why 136 is the printed wrong answer. The other wrong method to watch for is counting each layer once:  $100 + 10 + 2 + 8 = 120$ . Award nothing for (a) and say why: every layer has two sides. Give the method mark for a correct sum with one slip in the addition. "px" need not be written.*

## Paper B — Medium

30 MARKS • 35 MINUTES

- 1** blue: **rgb(0, 0, 255)** [1] [4]  
 maroon: **rgb(128, 0, 0)** [1]  
 lime: **#00FF00** [1]  
 yellow: **rgb(255, 255, 0)** [1]
- The 80 = 128 conversion separates a student who understands base 16 from one who guesses. rgb(80, 0, 0) for maroon is the tell-tale wrong answer: each pair read as ordinary decimal. Accept rgb written without spaces and lower-case hex. A hex code without the # loses the mark: the # is part of the code.*
- 2** The second option, color: red; [1] [1]
- The silent-failure trap. colour: red; is the tempting one: the browser ignores it and the text stays black, with no error message. color = red; has no colon; text-color is not a property.*
- 3** The second option, <a href="menu.html">Menu</a> [1] [1]
- Option 1 is href and src swapped; option 3 is an img with an href and a closing tag it must not have; option 4 opens with a and closes with link. One mark, no half marks.*
- 4** top: **5** [1] [4]  
 right: **10** [1]  
 bottom: **15** [1]  
 left: **20** [1]
- Four values go clockwise from the top: 12, 3, 6, 9 o'clock. The usual wrong answer reads them left to right as top, left, bottom, right (left 10, right 20). That still gets top and bottom right, so award those two marks only.*

- 5 Model answer: [4]
- ```
<ul>
  <li><a href="index.html">Home</a></li>
  <li><a href="menu.html">Menu</a></li>
  <li><a href="contact.html">Contact</a></li>
</ul>
```
- ul** opening and closing tags (an **ol** is a numbered list, and the question said unordered) [1]
 three **li** elements, each opened and closed [1]
 each **li** holds an **a** with **href** set to exactly `index.html`, `menu.html` and `contact.html` [1]
 link texts Home, Menu, Contact between the opening and closing a tags [1]
- This is the navigation bar of nearly every website, and the mark scheme is about nesting: ul, then li, then a. A link placed beside the li instead of inside it loses the second mark. A wrong file name loses the third. All three on one line, capitals, single quotes, any indentation: all fine.*
- 6 Line 2: `<a src="ruins.html">` should be `<a href="ruins.html">` [1] [3]
 Line 2: the **p** is never closed: add `</p>` after the link [1]
 Line 3: the **img** has no **alt**: add `alt="..."` describing the picture [1]
- One mark per fault, and the fault must be both found and corrected. The line number or a quotation of the line are both accepted. The missing `</p>` is the one students miss, because the browser forgives it; the exam does not. An `alt` of "image", "photo" or "temple.jpg" is not a correction.*
- 7 **h1** text colour: **green** [1] [4]
h1 alignment: **right** [1]
p text colour: **black** [1]
p alignment: **left** [1]
- Built around silent failure. colour and centre are British spellings, so the browser threw the whole p rule away: the paragraph stays black and left-aligned. A student who answers "blue, middle" has read what the CSS meant; the browser reads what it says. Mark the h1 pair independently of the p pair. The h1's right alignment does not reach the p.*
- 8 (a) $120 + 2 \times 20 + 2 \times 5 + 2 \times 10$ [1] [4]
 $= 120 + 40 + 10 + 20 = 190 \text{ px}$ [1]
 (b) $120 + 40 + 10 = 170 \text{ px}$ (margin left out) [1]
 (c) $(200 - 170) \div 2 = 15 \text{ px}$ [1]
- Counting each layer once gives 155 for (a): no marks, because every layer has two sides. In (b), 190 means the margin was included in the visible box. In (c), an answer of 30 means the margin was treated as one-sided; the working $200 - 170 = 30$ is worth nothing without the halving.*
- 9 (a) **padding** [1] [3]
 it is the space *inside* the border, and the background colour fills it [1]
 (b) **margin**: it is outside the border, so it pushes the neighbour away and is never coloured [1]
- The two words are swapped in most wrong scripts. Award (b) independently of (a). A student who says "border" for (a) has noticed the gap but not the layer: no mark. For (a) the reason must say inside or mention the background colour.*

- 10 Model answer: font-family: Georgia, serif; [2]
the property **font-family** with Georgia as the first choice [1]
a **generic family** as the safety net: serif (sans-serif or monospace are also accepted) [1]

The generic family is the mark most students drop: if the visitor does not have Georgia, the browser needs something to fall back on. Accept quote marks round Georgia, and "Georgia, 'Times New Roman', serif". font: on its own is a different property and earns nothing. A missing semicolon on its own is not penalised here: it is the last thing written.

Paper C — Hard

30 MARKS · 45 MINUTES

- 1 A: **margin** [1] [4]
B: **padding** [1]
C: **content** (accept "text") [1]
D: **border** [1]

No reminder box on this paper, so this is pure recall of the four layers. Same diagram as Paper A with the arrows pointing at different layers, so a student who memorised A's answers by position scores nothing. Margin and padding swapped is the usual error: withhold both marks.

- 2 (a) **rgb(255, 128, 0)** (FF = 255, 80 = $8 \times 16 = 128$, 00 = 0) [1] [3]
(b) **#000080** (128 = 8×16 , so 80) [1]
(c) **rgb(192, 192, 192)** (C0 = $12 \times 16 = 192$) [1]

Two wrong methods to look for. Reading a pair as ordinary decimal gives rgb(255, 80, 0) for (a). Pasting decimal into hex gives #0000128 for (b), seven digits. Accept lower-case hex; the # is required. No calculator is needed: C is twelve, and 12×16 is the only multiplication in the question.

- 3 Model answer: [3]
`<a href="home.html"></a>`
the **img** sits *inside* the **a**, between its opening and closing tags [1]
href on the **a** (home.html) and **src** on the **img** (logo.png), not swapped [1]
an **alt** that describes the picture: a green kite on a white circle [1]

Targets href/src and alt. alt="logo" or alt="logo.png" scores nothing: it names the file, it does not describe what is in it. An img placed beside the a (not inside it) loses the first mark only. A separate text link instead of the picture contradicts the question: first mark withheld. Single quotes, capitals, /> on the img: all accepted.

- 4 Line 5: the head is never closed: add `</head>` before `<body>` [1] [5]
Line 7: `<h4>` straight after the `h1` skips levels: it should be `<h2>` (structure, not size) [1]
Line 8: the **p** is never closed: add `</p>` after "January." (anywhere before the `ul`) [1]
Line 13: **href** on an image: it must be `src="tulip.jpg"` [1]
Line 13: `alt="image"` describes nothing: needs a real description, e.g. "Red tulips in the Glass House" [1]

One mark per fault, found (line number or a quotation of the line) and corrected. Two of the five, the h4 and the alt, would never make a browser complain, so a student who tests by "does it show?" finds three. An h3 for line 7 is not accepted: it still skips h2. A student who spots "two faults on line 13" but fixes one earns one mark. The page was checked with a real parser: these five are the only faults it has.

5 Fixed CSS:

[5]

```
h1 {
  color: white;
  background-color: #008000;
  text-align: center;
}
p {
  font-family: Georgia, serif;
  color: darkred;
}
```

(a) Line 2: **colour** should be color: white; [1]

Line 3: #00800 has five digits and a hex colour needs six: #008000 (any valid green) [1]

Line 4: centre should be text-align: center; [1]

Line 7: the semicolon after Georgia is missing: font-family: Georgia; (or Georgia, serif;) [1]

(b) The missing semicolon makes the browser read lines 7 and 8 as **one** declaration it cannot understand, so it throws away both, including the correct color: darkred; [1]

Both parts are one idea: CSS ignores whatever it cannot read and says nothing. In (a), the colour for line 3 may be any valid colour that is green: #080 or lime are acceptable; seven digits are not. For (b), "because there is no semicolon" alone scores nothing: the mark is for saying that line 8 was swallowed along with line 7.

6 Model answer:

[5]

```
<h1>Masala Dosa</h1>

<h2>Ingredients</h2>
<ul>
  <li>Rice</li>
  <li>Urad dal</li>
  <li>Potato</li>
</ul>
<h2>Method</h2>
<ol>
  <li>Soak</li>
  <li>Grind</li>
  <li>Cook</li>
</ol>
```

h1 then two **h2**, so the outline has no skipped level [1]

img with src="dosa.jpg" and an alt that describes the dosa (not "image", "photo" or "dosa.jpg") [1]

Ingredients: a **ul** holding three li (Rice, Urad dal, Potato) [1]

Method: an **ol** holding three li in the order Soak, Grind, Cook [1]

every tag opened is closed and nested correctly, only li inside the lists, no typed numbers, and no closing tag on the img [1]

Almost all of these marks are for structure, not wording. A heading level other than h2 for the two sections is fine only if the outline stays unbroken (h1, h2, h2 is the only sensible answer). The last mark is the one most often lost: a forgotten or . Extra doctype, html, head and body tags are harmless if they are correct. If the picture is placed elsewhere in the body, still award the list and structure marks.

- 7 (a) padding left and right: $25 + 25 = 50$ [1] [5]
 $150 + 50 + 8 + 12 = 220$ px (width, padding, border $4 + 4$, margin $12 + 0$) [1]
 (b) $300 - (50 + 8 + 12) = 300 - 70 = 230$ px [1]
 (c) vertical pairs: padding $10 + 10 = 20$, border $4 + 4 = 8$, margin $8 + 8 = 16$ [1]
 $60 + 20 + 8 + 16 = 104$ px [1]
- Two shorthands, two directions. padding: 10px 25px is vertical first, so left and right are 25; reading it the other way gives 190 for (a). margin: 8px 0 8px 12px runs top, right, bottom, left, so the right margin is 0; giving it 12 on both sides gives 232. Neither is close to 220, so the wrong methods show themselves. In (c) the left and right values are irrelevant: a student who adds 25 or 12 has not understood which direction is being measured. Give method marks for correct pairs even if the final addition slips.*

Paper D — Hard

30 MARKS • 45 MINUTES

- 1 Model answer: [3]

```
<!DOCTYPE html>
<html>
<head>
  <title>Jaipur Kites</title>
</head>
<body>
  <h1>Jaipur Kites</h1>
  <p>Festival on 14 January.</p>
</body>
</html>
```

<!DOCTYPE html> and the **html** element wrapped round everything [1]
head containing the **title** Jaipur Kites [1]
body containing the h1 and the p, correctly nested and closed [1]
- The most reliable three marks on the paper and the most often thrown away. Usual losses: the title written in the body (so it never appears on the tab), the heading inside the head, or no closing html. The doctype may be in any case, but if it is missing, withhold the first mark.*
- 2 Model answer: [3]

```
p { font-family: Verdana, sans-serif; color: #008000; text-align: right; }
```

a rule of the right shape: selector p, curly braces, property, colon, value, semicolon [1]
font-family with Verdana first and a generic family last (sans-serif) [1]
color: #008000; (0, 128, 0 is 00, 80, 00) and **text-align: right;** [1]
- The conversion is the point: 128 is $8 \times 16 = 80$ in hex. #00FF00 is lime (255); #00128 and #0012800 are decimal pasted into hex. The question demands a hex code, so "green" scores nothing for the colour. Accept lower-case hex. colour and right-align lose the third mark.*
- 3 (a) everything except the width: $2 \times 15 + 2 \times 5 + 2 \times 10 = 30 + 10 + 20 = 60$ [1] [3]
 $400 - 60 = 340$ px [1]
 (b) a border needs a **style** as well as a width: 5px red has none, so no border is drawn (border: 5px solid red; fixes it) [1]
- The question asks the box to fit, not to be built. A student who ADDS the 60 and answers 460 has built the box: tell them to add $340 + 60$ and see. Subtracting only one layer gives 385 (padding once) or 395 (border once). In (b) "it needs a colour" is wrong: the colour is there. The word the mark scheme wants is style (or solid).*

- 4 (a) Model answer: [5]
- ```

 Home
 Shop
 Contact

```
- ul** with three **li**, each opened and closed [1]  
 each **li** holds an **a** with href exactly index.html, shop.html and contact.html [1]  
 link texts Home, Shop, Contact between the a tags [1]  
 (b) 3 pages × 3 links = **9** [1]  
 (c) **3** files: all three pages carry a link to shop.html, the shop page itself included [1]
- Part (b) is the answer to the cliffhanger at the end of lesson 6: how many links does a three-page site really need? Nine, and every one is typed by hand, which is why renaming one page means editing every page. In (c), 2 means the page forgot it links to itself; 1 means the other two pages were forgotten. The number needs the reason to score.*
- 5 (a)  $80 + 2 \times 10 + 2 \times 2 + 2 \times 6 = 80 + 20 + 4 + 12$  [1]  
 = **116 px** [1]  
 (b)  $3 \times 116 =$  **348 px** [1]  
 348 is less than 360, so they **do fit** (12 px to spare) [1]  
 (c)  $80 + 20 + 4 + 20 = 124$ , and  $3 \times 124 =$  **372 px** [1]  
 372 is more than 360, so they **do not fit**: 12 px too wide [1]
- The wrong method prints itself:  $3 \times 80 = 240$  looks like 120 px to spare, when the real figure is 12. The question is also built so that forgetting the border flips part (c): without it each box is  $80 + 20 + 20 = 120$ , three make exactly 360, and the student answers "fits". Award (b) and (c) independently and allow follow-through from a wrong (a), as long as the comparison with 360 is made.*
- 6 (a) Line 4: `<a src="virupaksha.html">` should be `<a href="virupaksha.html">` [1]  
 Line 5: a `<p>` sits directly inside the `ul`: it must be `<li>Vittala Temple</li>` [1]  
 Line 7: `alt="stone-chariot.jpg"` is the file name: describe the picture, e.g. "Stone chariot at Vittala Temple" [1]  
 (b) a visitor who cannot see the picture would hear the file name, which says nothing about what is in it [1]
- Three different slips from three lessons: href/src, a list with something other than li in it, and alt text that is only the file name. The page checks out clean once these are fixed. A student who deletes line 5 instead of changing it has lost content: no mark. For (b), "for search engines" on its own is not the reason; it has to be about the visitor who cannot see.*
- 7 (i) Padding is the space **inside** the border (the background fills it); margin is the space **outside**. They are different layers [1]  
 Padding only makes the coloured box bigger: the gap between the two boxes' edges has not changed [1]  
 The correct property is **margin: 20px;** [1]  
 (ii) A border has thickness, so it **adds** to the size [1]  
 It adds on **both** sides:  $2 \times 4 = 8$  [1]  
 $120 + 20 + 8 =$  **148 px** (not 140) [1]
- The standing question type: two wrong statements in another student's handwriting. Marking someone else's error is far easier to face than being told you made it, and it is the same thinking. Both statements are the topic's two big traps in confident prose. If he can name the mistake in Divya's and Neel's notes, he is unlikely to write them in his own. For (ii), 148 with no reason scores one mark; the reasoning is what is being tested.*